

12.112 Matgeo

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Question

If the entries in each column of a square matrix $\mathbf{M}$ add up to 1, then an eigenvalue is

- ① 4
- ② 3
- ③ 2
- ④ 1

Solution

The question states that the sum of all elements in each column of the square matrix $\mathbf{M}$ is 1 then we have to find a number which is an eigen value of such matrix

Solution

If we take a matrix

$$\mathbf{v} = [1, 1, \dots, 1]^T \quad (1)$$

Then , we can say that

$$\mathbf{M}\mathbf{v} = \mathbf{v} \quad (2)$$

Solution

From the above equation we can say that one of the eigen value for **M** is 1
Hence option 4 is true